Testing Inheritance Patterns

Biology 2e chapter 12 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Each kernel develops from a fertilization event. Use documented parent genotypes and a defined visible trait before predicting offspring ratios. Sample size and chance affect observed counts.

Investigation question

How closely do observed kernel traits match an inheritance prediction?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

The model represents a documented single-locus Pp x Pp cross: P is dominant for purple pigment in this simplified trait. Before counting, predict the two phenotype proportions. Classify all 48 sampled kernels on each ear.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: kernels

Condition / sample	Repeat	Source	Purple	Yellow
Ear A	1			
Ear B	1			
Ear C	1			

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Compare observed and expected counts under a 3:1 prediction.

Calculate chi-square: sum (observed - expected)^2 / expected. What does that statistic measure?

Would a small deviation alone disprove Mendelian segregation?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Generated kernel populations and original artwork, not photographs or supplier data. Real corn pigment genetics can involve more than one locus; apply the prediction only to a documented suitable cross.